

CRESTON, IA, USA

WMO: 725474

Lat: 41.020N

Lon: 94.364W

Elev: 394

StdP: 96.68

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04915

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.4	-17.5	-24.8	0.4	-20.1	-22.2	0.5	-16.8	14.2	-4.9	12.8	-4.1	4.0	300	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	33.0	24.8	32.3	24.3	30.8	23.5	26.6	31.5	25.5	30.0	24.5	28.9	5.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	21.2	29.7	24.0	19.8	28.7	22.9	18.5	27.4	85.5	31.5	80.7	29.9	76.3	29.0	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.3	11.0	9.4	-24.5	35.7	3.0	2.1	-26.6	37.2	-28.4	38.5	-30.0	39.7	-32.2	41.2
			-24.8	27.6	2.9	1.1	-26.8	28.4	-28.5	29.1	-30.2	29.8	-32.3	30.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	-4.5	-2.8	3.8	10.6	16.7	21.8	24.0	22.8	18.5	11.5	4.2	-2.8
	DBStd	11.44	6.81	6.91	6.60	5.37	4.56	3.45	3.26	3.15	4.55	5.36	5.86	6.48
	HDD10.0	1718	451	358	213	56	4	0	0	0	2	47	189	399
	HDD18.3	3445	709	591	452	238	85	10	2	4	53	221	425	656
	CDD10.0	1854	0	1	20	72	212	353	435	396	258	92	14	1
	CDD18.3	539	0	0	1	4	35	112	179	142	58	8	0	0
	CDH23.3	4938	0	0	6	64	316	996	1758	1212	513	72	2	0
	CDH26.7	1754	0	0	1	13	87	333	708	438	161	13	0	0
Wind	WSAvg	4.6	5.0	4.9	5.3	5.7	5.1	4.3	3.5	3.3	3.8	4.5	4.9	4.7
Precipitation	PrecAvg	902	25	29	57	93	126	130	115	101	84	59	49	32
	PrecMax	1289	50	65	143	187	247	332	312	281	298	147	129	71
	PrecMin	654	10	5	2	35	36	22	30	31	26	6	1	3
	PrecStd	192	10	14	36	38	52	75	66	66	61	41	33	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	17.6	23.9	28.6	32.3	32.9	36.2	35.2	32.5	28.7	22.6	15.9
		MCWB	7.5	11.3	15.5	17.8	22.4	24.0	24.8	24.6	22.8	19.3	14.6	11.4
	2%	DB	9.8	13.0	20.3	25.1	28.9	32.1	33.8	32.7	30.9	25.2	18.8	11.4
		MCWB	6.1	8.8	13.5	16.3	21.0	23.8	25.6	24.8	22.4	17.0	13.4	8.5
	5%	DB	7.1	9.9	17.3	22.5	27.0	30.2	32.4	31.1	28.1	22.5	15.9	8.6
		MCWB	4.2	6.3	12.3	15.2	19.5	23.0	25.0	24.0	21.2	16.3	10.8	5.8
	10%	DB	3.9	7.0	13.9	19.9	24.8	28.7	30.9	28.9	26.3	20.1	12.8	6.0
		MCWB	1.8	4.1	9.9	13.6	18.4	22.2	24.2	23.2	19.7	15.1	8.8	3.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	13.1	17.5	19.7	24.5	26.5	28.3	27.8	25.0	22.1	16.3	13.0
		MCDB	12.2	15.1	21.3	25.9	28.6	31.2	33.3	31.5	30.1	26.5	20.2	14.9
	2%	WB	6.3	9.2	15.2	17.8	22.1	25.1	26.9	26.1	23.8	19.8	14.0	9.2
		MCDB	9.7	13.7	18.2	23.3	27.3	29.6	32.2	30.5	28.6	23.4	17.5	10.6
	5%	WB	4.1	6.5	12.8	16.1	20.8	24.1	25.9	25.0	22.7	17.9	11.8	5.8
		MCDB	6.9	9.5	16.6	21.2	25.6	28.7	30.7	29.2	26.9	20.4	15.1	8.1
	10%	WB	2.1	3.9	10.0	14.4	19.5	23.1	24.9	23.9	21.3	15.7	9.4	3.6
		MCDB	4.0	6.5	14.3	18.7	23.4	27.5	29.3	28.2	24.9	19.5	13.1	5.7
Mean Daily Temperature Range	5% DB	MDBR	9.6	9.8	10.8	11.6	10.7	10.4	10.6	10.7	12.1	11.3	10.4	8.9
		MCDBR	13.0	13.7	15.3	15.6	12.9	11.6	11.9	12.6	13.3	13.7	14.2	12.0
		MCWBR	9.7	9.5	9.3	9.1	6.7	5.8	5.7	6.2	6.6	7.4	9.0	8.9
	5% WB	MCDBR	12.3	12.8	13.3	13.5	11.1	10.5	10.5	10.8	11.4	11.2	12.6	10.9
MCWBR		9.4	9.5	8.9	8.8	6.4	6.4	6.2	6.6	6.4	7.8	9.2	8.8	
Clear-Sky Solar Irradiance	taub	0.287	0.310	0.326	0.380	0.407	0.413	0.427	0.418	0.380	0.334	0.295	0.283	
	taud	2.427	2.374	2.406	2.272	2.269	2.297	2.248	2.289	2.346	2.468	2.514	2.457	
	Ebn at Noon	875	901	926	891	871	864	847	844	855	864	863	850	
	Edh at Noon	79	97	105	129	133	130	135	126	111	88	72	71	
All-Sky Solar Radiation	RadAvg	2.01	2.96	3.82	4.82	5.69	6.32	6.36	5.53	4.66	3.18	2.19	1.69	
	RadStd	0.18	0.19	0.34	0.33	0.46	0.56	0.47	0.34	0.32	0.37	0.18	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page